COMPONENT COHOMOLOGY AGREES WITH TOTAL-SITE COHOMOLOGY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sources/stacks-spaces-simplicial/pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The label lemma-sanity-check-modules names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The counted proof is a complete special-case application: every hypothesis of the cited external lemma is verified or definitional inside the counted span.

Allowed standard prerequisites (editorial).

- Pullback does not change cohomology. Cohomology on Sites chapter, \label{sites-cohomology-lemma-pullback-same-cohomology}. Lines 2226-2228: the external lemma whose hypotheses are verified in the counted proof.
- Component functors are continuous and cocontinuous. Same chapter, \label{lemma-restriction-to-components-site}. Lines 2222-2224: supplies the hypothesis for the external lemma.

Source reference. Original lemma 09WC: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 797–797. Prerequisite pointer invoked at proof line(s) 2218; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09WD: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 844–844. Prerequisite pointer invoked at proof line(s) 2220; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original lemma 09WG: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 970–970. Prerequisite pointer invoked at proof line(s) 2222; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Date: stacks-project:spaces-simplicial.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0DD8: sources/stacks-chow/tags.txt, source lines 16758–16758. Prerequisite pointer invoked at proof line(s) 2226; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 09WE: sources/stacks-spaces-simplicial/pinned-original-source.tex, source lines 888–888. Prerequisite pointer invoked at statement line(s); it names the Situation supplying the running notation of the chapter, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *In Situation 09WE let $\mathcal{O}$ be a sheaf of rings. Let $U \in \mathrm{Ob}(\mathcal{C}_n)$. Let $\mathcal{F} \in Mod(\mathcal{O})$. Then $H^p(U, \mathcal{F}) = H^p(U, g_n^*\mathcal{F})$ where on the left hand side U is viewed as an object of $\mathcal{C}_{total}$.*

Proof. Observe that "U viewed as object of $\mathcal{C}_{total}$" is explained by the construction of $\mathcal{C}_{total}$ in Lemma 09WC in case (A) and Lemma 09WD in case (B). In both cases the functor $\mathcal{C}_n \to \mathcal{C}$ is continuous and cocontinuous, see Lemma 09WG, and $g_n^{-1}\mathcal{O} = \mathcal{O}_n$ by definition. Hence the result is a special case of Cohomology on Sites, Lemma 0DD8. □